

Review of “FedDecay: Balancing Model Performance and Rapid Personalization in Federated Learning with Learning Rate Scheduling”

Summary

This manuscript presents *FedDecay*, a modification of standard Federated Learning (FL) algorithms that introduces within-round exponential learning rate decay during local updates. This technique enables tuning the trade-off between global model generalization and local model personalization. The method is framed as a generalization of FedAvg and FOMAML, and is theoretically supported through a gradient decomposition analysis.

The authors validate their approach through experiments on standard datasets spanning vision, text, and graph data. The results suggest that FedDecay consistently improves test performance for both new and existing users while incurring minimal additional computational or memory cost compared to baseline FL methods.

General Comments

The paper addresses an important challenge in federated learning: balancing shared model generality with the ability to personalize to heterogeneous users. The proposed solution is relatively simple, theoretically grounded, and does not require architectural changes or significant overhead. The paper is generally well-written and structured, and the experimental results demonstrate solid performance improvements. Overall, the approach is a meaningful and practical extension to existing FL methods.

Technical Comments

1. The paper discusses computational efficiency but does not evaluate whether learning rate decay impacts convergence in terms of the number of communication rounds. Including this analysis would offer further insight into the method’s potential efficiency benefits.
2. FedDecay appears most effective under moderate data heterogeneity. To enhance practical usability, it would be helpful to offer a heuristic or metric to guide users in choosing between FedDecay and more personalized alternatives.
3. While FedDecay is efficient, it relies on grid search to tune the decay parameter β . It would be helpful to discuss:
 - The sensitivity of performance to different β values;
 - Whether coarse tuning (e.g., using $\{0.5, 0.7, 0.9\}$) is generally sufficient;
 - Any observed patterns that might assist users in selecting β based on data properties.

Recommendation

Accept with Minor Revisions. The paper offers a valuable and practical contribution to federated learning. Addressing the comments above would improve the clarity and applicability of the method in real-world settings.